

$(t_0 \sigma_0 Q (FS_0 \dots (\text{stack (frame } e_0 l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack (frame } e_1 l) F \dots) FS_1 \dots))$

where $(\text{not value?}[\![e]\!]), (\sigma_1 e_1) = \Downarrow \text{base}$

$(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack (frame } E[(\text{async/lambda } (x \dots_1) e) v \dots_1] l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack (frame } E[(\text{task } x_{\text{async}})] l) F \dots) FS_1 \dots))$

where (pt

$(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack (frame } v l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack } F \dots) FS_1 \dots))$

where $(\text{async? } l), x_{\text{async}} = l, \text{none} = \text{task-tree-run}$

$(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack (frame (throw } v_{\text{err}}) l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack } F \dots) FS_1 \dots))$

where $(\text{async? } l), x_{\text{async}} = l, v_{\text{ol}}$

$(t_0 \sigma_0 Q (FS_0 \dots (\text{stack parent-frame } F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack } F \dots) FS_1 \dots))$

[task-ret

where $(\text{async? } l), x_{\text{async}} = l, (\text{some } x_{\text{child}}) = \text{task-tree-running-child}[\![\sigma_0, x_{\text{async}}]\!], \text{parent}$

$(t_0 \sigma Q (FS_0 \dots (\text{stack (frame } E[(\text{await (task } x_{\text{async}})]) l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma Q (FS_0 \dots (\text{stack (frame } E[v] l) F \dots) FS_1 \dots))$

where (done v) = task-sta

$(t_0 \sigma Q (FS_0 \dots (\text{stack (frame } E[(\text{await (task } x_{\text{async}})]) l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma Q (FS_0 \dots (\text{stack (frame } E[(\text{throw } v)] l) F \dots) FS_1 \dots))$

where (failed v) = task-sta

$(t_0 \sigma_0 Q (FS_0 \dots (\text{stack current-frame } F \dots) \longrightarrow$
 $FS_1 \dots))$

$(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack } F \dots) FS_1 \dots))$

where $(\text{async? } l), v_{\text{obj}} = \text{lookup}[\![\sigma_0, x_{\text{async}}]\!], (\text{pending } _ \dots) = \text{task-state}[\![v_{\text{obj}}]\!], \text{current-frame} = (\text{frame } E[(\text{a$

$(t_0 \sigma_0 Q (FS_0 \dots (\text{stack (frame } E[(\text{cancel (task } x_{\text{async}})]) l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack (frame } E[(\text{void})] l) F \dots) FS_1 \dots))$

$(t_0 \sigma Q (FS_0 \dots (\text{stack (frame } E[(\text{cancelled?})] l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma Q (FS_0 \dots (\text{stack (frame } E[\text{boolean}] l) F \dots) FS_1 \dots))$

where (a

$(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack (frame } E[(\text{os/fio natural } v)] l) F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma_1 Q_1$

$(FS_0 \dots (\text{stack (frame } E[(\text{task } x_{\text{async}})] l) F \dots) FS_1 \dots))$

where $(\text{ptr } x_{\text{async}}) = \text{malloc}[\![\sigma_0]\!], \sigma_1 = \text{ext1}[\![\sigma_0, (x_{\text{async}} \text{ new-task}[\![\!]\!])]\!], Q_1 = \text{q-push}[\![Q_0, (\text{frame (os/resolve (task } x_{\text{async}}) \text{ lib:}\Sigma[\![t_0, \text{natural}]\!]) v]$

$(t_0 \sigma_0 Q_0 \longrightarrow$
 $(FS_0 \dots (\text{stack (frame } E[(\text{os/resolve (task } x_{\text{async}}) t_{\text{resolve}} v)] l) F \dots)$
 $FS_1 \dots))$

$(t_1 \sigma_1 Q_1 (FS_0 \dots (\text{stack } F \dots) FS_1 \dots))$

where $(\geq t_0 t_{\text{resolve}}), v_{\text{ol}}$

$(t_0 \sigma Q_0 (FS_0 \dots (\text{stack current-frame } F \dots) FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma Q_1 (FS_0 \dots (\text{stack } F \dots) FS_1 \dots))$

[os

where $(< t_0 t_{\text{resolve}}), \text{current-frame} = (\text{frame } E[(\text{os/re$

$(t_0 \sigma Q_0 (FS_{\text{main}} FS_0 \dots (\text{stack } FS_1 \dots)) \longrightarrow$
 $(t_1 \sigma Q_1 (FS_{\text{main}} FS_0 \dots (\text{stack } F_{\text{head}}) FS_1 \dots))$

where $\text{all-busy?}[\![FS_0, \dots]\!], (F_{\text{head}} Q_1) = \text{q-p$

$(t_0 \sigma Q ((\text{stack (frame } E[(\text{block (task } x_{\text{async}})]) l) FS_{\text{rest}} \dots)) \longrightarrow$
 $(t_1 \sigma Q ((\text{stack (frame } E[v] l) FS_{\text{rest}} \dots))$

where $(\text{sync? } l), (\text{not any-busy?}[\![FS_{\text{rest}}, \dots]\!]), (\text{done } v) = \text{task-sta$